

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 50

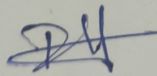
Annexure A

Scenario Based Assignment

Code of Conduct:

I Shaik Fahd (192565026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



20/18-12345678912
791
42

1. Scenario

A hospital is designing an AI system to automatically generate monthly duty rosters for its nursing staff. The system must satisfy multiple constraints: no nurse should be scheduled for two overlapping shifts, a limited number of nurses are qualified for the ICU, certain senior nurses must supervise night shifts, nurses have individual leave and availability constraints, and special accommodations are needed for part-time staff. The complexity increases as the number of nurses and wards grows.

Tasks:

Formulate the problem as a Constraint Satisfaction Problem (CSP).

- i. Clearly define variables, domains, and constraints with explanation.
- ii. Explain how backtracking search works in this scenario.
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.
- iv. Analyze real-world challenges and justify the best strategy for scalability.

2. Scenario

A mobile gaming company is developing an AI opponent for a chess app aimed at casual players. The AI must compete against human players, make decisions within strict time limits (a few seconds per move), evaluate complex board positions with many possible moves, and the decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem.
- ii. Analyze the computational challenges of large game trees.
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.
- iv. Design an evaluation function and justify the factors considered.
- v. Recommend a strategy for balancing optimality and real-time performance.

3. Scenario

A disaster-response agency deploys AI-guided drones to airlift medical supplies to villages cut off by a wildfire. The environment is highly dynamic - access roads and forest trails may become impassable without warning, and smoke conditions affect travel cost. The drone has access to: a heuristic estimate (straight-line distance to each village), partial real-time updates about newly blocked trails, and variable movement costs depending on terrain and visibility. Due to the urgency of the situation, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic.
- iii. Analyze the impact of heuristic accuracy on both algorithms.
- iv. Discuss how dynamic changes (newly blocked trails) affect search performance.
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

extremely large, weather patterns change dynamically, and the system may get stuck in locally optimal valve configurations.

Tasks:

Formulate this as an optimization problem and explain why traditional search is inefficient.

- i. Explain how Hill Climbing would work in this scenario and identify its limitations.
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
- iv. Recommend the best approach and justify your choice based on scalability and adaptability.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 + 18 + 23 + 18 + 12$$

(a1)

PP

1. Scenario - Hospital Nurse Duty Roster

The hospital duty roster problem can be formulated as a constraint satisfaction problem consisting of variables, domains and constraints

Constraints :-

1) A nurse cannot be assigned to two overlapping shifts.

2) Only ICU-qualified nurses can be assigned to ICU duties.

3) Senior nurses must be assigned to require night shifts.

4) A nurse cannot be scheduled during approved leave.

5) Nurse cannot be scheduled during approved leave.

6) Nurse availability must be respected

Basic Process:

Start $\rightarrow$ Select shift $\rightarrow$ Assign Nurse
 $\rightarrow$ Check constraints $\rightarrow$ Valid? $\rightarrow$ continue;
Back track $\rightarrow$ compute roster

Constraint propagation / forward checking :-

Forward checking improves back tracking by removing impossible values from the domains of unassigned variables.

This detects conflicts early instead of waiting the complete roster is built.

Therefore, fewer combinations need to be explored and the search becomes faster.

For very large hospitals, the problem can additionally be divided by wards or departments and solved using csp solvers

3. Scenario - AI chess Opponent:-

Minimax Algorithm:-

Chess can be represented as a game tree where each node represents a board position and each edge represents a possible move.

The AI is the MAX player, trying to maximize its score, while the human opponent is the MIN player, trying to minimize the AI's score.

Algorithm:-

Current position



possible moves



opponent responses



further moves → Evaluation.

15
tational challenges:-

Chess has a very large game tree
each position can have many legal
moves. Exploring every possible sequence until
the end of the game.

Time complexity = $O(b^d)$.

Alpha - Beta pruning:-

Alpha-Beta pruning improves minimax
by eliminating branches that cannot influence
the final decision.

Alpha (α) :- Best Max value guarantee.

Beta (β) :- Best Min value guarantee.

When $\alpha \geq \beta$, the remaining branch
can be ignored.

3. Scenario - Disaster-Response Drone Routing.

Greedy Best-First search

Greedy Best-First Search selects the node with the smallest heuristic value:

$$f(n) = h(n)$$

Here $h(n)$ can be the straight-line distance from the drone's current location to the village.

ii) A^* Search

A^* combines the actual cost and estimated remaining cost

$$f(n) = g(n) + h(n)$$

where

$g(n)$ = Actual cost from the starting point to node n .

$h(n)$ = estimated cost from n to destination

$f(n)$ = estimated total route cost

Recommended Algorithm:-

A* is most suitable primary algorithm because the drone must consider both route cost and estimated distance while maintaining good path quality.

However because the environment changes rapidly, A* should be combined with

- * Real-time map updates
- * Risk and terrain costs
- * A suitable heuristic
- * Frequent route re-planning
- * A time limit for emergency decisions.